

GABLS1: setup and results

Part II: GABLS1

14/15 completed exports admitted to the scientific comparison. Checked 2026-09-21T04:25:39.448Z.

Grids: 12.5, 6.25, 3.125, 2, and 1 m isotropic; 32^3 , 64^3 , 128^3 , 200^3 , and 400^3 cells, all in the same 400 m cube. Small grids run first.

Each grid has exactly three configurations: WENO9 with no interior closure, WENO5 with no interior closure, and WENO9 with Smagorinsky. Smagorinsky $C_s=0.16$, $Pr=1$; paired seed 123 within each grid. These are controlled Breeze settings, not a reproduction of one named historical LES member.

Forcing: $U_g = 8$ m/s, $V_g = 0$, $f = 1.39e-4/s$. Initial potential temperature 265 K below 100 m and $265+0.01(z-100)$ K above. Initial temperature noise is uniform with 0.1 K peak-to-peak range below 50 m. Surface $\theta_{s,ref}=265-0.25t_{hours}$ K; $\theta_{ref}=263.5$ K is a separate buoyancy reference.

Wall: MOST exchange with $\kappa = 0.4$, $\beta_m = 4.8$, $\beta_h = 7.8$, roughness = 0.1 m for heat and momentum. Impermeable boundaries, free-slip upper lid. Vertical-velocity damping is zero below 300 m and ramps linearly to a 60 s relaxation time at 400 m. MOST uses a minimum wind of 0.01 m/s and caps z/L at 10, with a neutral fallback for nonpositive bulk Richardson number; their activity is diagnosed. Exact implementation, damping and timestep choices are recorded in production provenance.

Diagnostics: 541 instantaneous series records at 60 s intervals; 19 profile times with preceding 30 min means after instantaneous t_0 . Compare 7-8 h and 8-9 h. Final-hour profiles combine bins ending at 30600 and 32400 s; penultimate-hour bins end at 27000 and 28800 s.

Compare mean $u/v/\theta$, jet speed/height/turning, total-stress boundary-layer height, surface heat flux/ u_{star}/L , native-face $w_2/w_3/skewness$, resolved/SGS flux partitions and available TKE-budget terms. Missing SGS energy or budget terms remain unavailable; a residual is not numerical dissipation.

Beare et al. (2006), Boundary-Layer Meteorology 118, 247-272, doi:10.1007/s10546-004-2820-6. Official reduced data archive contains 423 files. Native coordinates and missing values are retained; some NERSC headers give 474-534 min for nominal 8-9 h, so exact window matching is not universal.

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Cluster agents own implementation, scheduling and audited exports. Desktop monitoring is scheduled every 10 min to collect newly completed cases and rebuild this document. Actual submitted jobs and completed-export evidence are recorded in the status files, separate from the authorized matrix.

Production submission records: 15/15 cases have assigned Slurm jobs. Submitted is distinct from running or scientifically complete. The live queue snapshot below also includes any smoke tests.

Implementation choices: Breeze uses anelastic dynamics and Float32 arithmetic; the reference archive is predominantly Boussinesq. Base pressure is 100000 Pa so the prescribed surface temperature equals surface potential temperature; the corresponding reference density differs from the specification's rounded 1.3223 kg/m³ by about 8 parts per million. Smagorinsky-Lilly uses $C_s=0.16$, $C_b=1$, $Pr=1$.

At $\Delta=12.5$ m, WENO9/Smagorinsky has strongly suppressed resolved turbulence: peak final-hour w_2 is $1.12e-06$ m²/s², $1.21e-05$ times the no-closure value. Maximum magnitudes of the final-hour mean resolved and SGS momentum-flux vectors are $1.34e-07$ and 0.0415 m²/s². Its h is 140.6 m with only 0.242 m standard deviation across final-hour 60 s samples. Similar h or mean wind can therefore coexist with very different resolved turbulence and transport partition; h alone cannot establish fidelity. Skewness in this weak-variance regime must not be interpreted as evidence of vigorous turbulent motions.

At $\Delta=12.5$ m without interior closure, WENO9 versus WENO5 gives final-hour h 195.6 versus 136.1 m and peak resolved w_2 0.0927 versus 0.0574 m²/s². This is a scheme sensitivity at fixed grid and seed, not evidence of resolution convergence.

At $\Delta=6.25$ m without interior closure, WENO9 versus WENO5 gives final-hour h 187.8 versus 150.9 m and peak resolved w_2 0.0976 versus 0.0656 m²/s². This is a scheme sensitivity at fixed grid and seed, not evidence of resolution convergence.

At $\Delta=3.12$ m without interior closure, WENO9 versus WENO5 gives final-hour h 184.4 versus 162.9 m and peak resolved w_2 0.1008 versus 0.0696 m²/s². This is a scheme sensitivity at fixed grid and seed, not evidence of resolution convergence.

At $\Delta=2$ m without interior closure, WENO9 versus WENO5 gives final-hour h 178.5 versus 159.2 m and peak resolved w_2 0.0993 versus 0.0690 m²/s². This is a scheme sensitivity at fixed grid and seed, not evidence of resolution convergence.

At $\Delta=1$ m without interior closure, WENO9 versus WENO5 gives final-hour h 165.3 versus 159.4 m and peak resolved w_2 0.0893 versus 0.0681 m²/s². This is a scheme sensitivity at fixed grid and seed, not evidence of resolution convergence.

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n032_weno9_none: peak final-hour resolved w_2 ($0.0927 \text{ m}^2/\text{s}^2$) exceeds the largest archived model peak at the same spacing ($0.0602 \text{ m}^2/\text{s}^2$). Model spread is a comparison baseline, not observational truth; stronger resolved turbulence alone does not establish better fidelity.

n064_weno9_none: peak final-hour resolved w_2 ($0.0976 \text{ m}^2/\text{s}^2$) exceeds the largest archived model peak at the same spacing ($0.0779 \text{ m}^2/\text{s}^2$). Model spread is a comparison baseline, not observational truth; stronger resolved turbulence alone does not establish better fidelity.

n128_weno9_none: peak final-hour resolved w_2 ($0.1008 \text{ m}^2/\text{s}^2$) exceeds the largest archived model peak at the same spacing ($0.0904 \text{ m}^2/\text{s}^2$). Model spread is a comparison baseline, not observational truth; stronger resolved turbulence alone does not establish better fidelity.

n400_weno9_none: peak final-hour resolved w_2 ($0.0893 \text{ m}^2/\text{s}^2$) exceeds the largest archived model peak at the same spacing ($0.0851 \text{ m}^2/\text{s}^2$). Model spread is a comparison baseline, not observational truth; stronger resolved turbulence alone does not establish better fidelity.

Resolution comparison for WENO9/none: reducing spacing from 12.5 to 1 m changes final-hour h from 195.6 to 165.3 m and peak resolved w_2 from 0.0927 to $0.08926 \text{ m}^2/\text{s}^2$. Surface heat flux changes from -15.52 to -8.05 W/m^2 . These compare the coarsest and finest completed grids for this configuration; similarity of selected quantities alone is not a convergence or fidelity test.

Resolution comparison for WENO5/none: reducing spacing from 12.5 to 1 m changes final-hour h from 136.1 to 159.4 m and peak resolved w_2 from 0.05744 to $0.06807 \text{ m}^2/\text{s}^2$. Surface heat flux changes from -11.79 to -7.47 W/m^2 . These compare the coarsest and finest completed grids for this configuration; similarity of selected quantities alone is not a convergence or fidelity test.

Resolution comparison for WENO9/smagorinsky: reducing spacing from 12.5 to 2 m changes final-hour h from 140.6 to 188.9 m and peak resolved w_2 from $1.12\text{e-}06$ to $0.09298 \text{ m}^2/\text{s}^2$. Surface heat flux changes from -9.17 to -12.22 W/m^2 . These compare the coarsest and finest completed grids for this configuration; similarity of selected quantities alone is not a convergence or fidelity test.

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For WENO9/Smagorinsky, the finest completed grid has $8.3\text{e}+04$ times the coarsest-grid peak final-hour resolved w^2 . At $\Delta=2$ m, maximum magnitudes of final-hour mean resolved and SGS momentum-flux vectors are 0.06499 and 0.04146 m^2/s^2 ; these maxima can occur at different heights and should not be treated as fractions of a common total. Final-hour h sample standard deviation is 4.63 m, and mean h changed by 3.31 m between 7–8 and 8–9 h. Read the variance evolution and flux profiles together: this resolution sensitivity does not establish statistical stationarity or identify a unique cause.

Final-hour values are descriptive results, not a fidelity ranking. Resolution comparisons must keep the scheme and closure fixed; archive ensemble membership changes with resolution.

The archive reports skewness directly; its averaging convention is not assumed identical to our ratio of time-averaged central moments. No archive third moment is reconstructed from independently averaged variance and skewness. Both Breeze skewness plots mask levels with final-hour variance at or below $1\text{e}-8$ m^2/s^2 ; the raw diagnostics remain available.

A fixed 1.0 m archive ensemble median (black dotted line) is repeated across resolution panels wherever a matching quantity is available, alongside same-grid archive spread. The 1.0 m archive has two models; pointwise member counts and source files are saved in `reference_data/fixed_1m_medians.json`. Linear totals and resolved TKE are computed per model before taking medians. No matching reference is fabricated for w^3 , time-mean stress magnitude, unmatched budget terms, wall-regime fractions, peak-variance time series or jet time series. The fixed reference aids resolution comparisons but is not an exact solution.

The 1.0 m IMUK archive contains conspicuously large upper-level SGS values, confirmed in the original files: SGS uw spans -0.872 to 0.869 m^2/s^2 and SGS shear production spans -0.283 to 0.793 m^2/s^3 . Tiny values approaching $1\text{e}-100$ abruptly become order-one values above roughly 330 m. This pattern suggests an archive exponent-format problem, but the intended values have not been established. Raw values and resulting median curves are retained, with affected flux/budget panels flagged. Do not interpret those upper-level excursions as a convergence target.

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Corrected production (v2) replaces the original campaign after a diagnostic index restriction truncated total-flux/stress profiles and exposed a boundary-layer-height contour-index error. The corrected source preserves the original physics and all three numerical configurations. Expanded CPU checks passed 278 kernel and 426 writer checks; GPU checks passed 268 kernel and 422 writer checks, including every raw profile extent. All 15 cases were resubmitted as arrays 7099-7103, beginning with 32³. Original outputs remain excluded from scientific comparisons and preserved as failure evidence. The collector and single cluster exporter now target v2.

The original 400³ WENO9/no-closure and WENO9/Smagorinsky attempts failed during startup and remain excluded. Their fixed initial timestep of 0.5 s implied an advective CFL of 4 for 8 m/s wind and 1 m cells. The separate startup_dt_v1 snapshot changes only initial dt to $\min(0.5, 0.5 \cdot dx/8)$, or 0.0625 s at 1 m. Both variants passed 72 s GPU tests with finite saved bounds: 852/832 iterations and final actual CFL 0.700088/0.700007. The unchanged wizard gave an effective first dt of 0.06875 s. Replacement jobs 7120_1 (WENO9/no closure) and 7120_2 (WENO9/Smagorinsky) were submitted with a one-job array limit; 7120_1 is running and 7120_2 queued at this snapshot. Original WENO5 job 7103_2 continues without duplication. Original failed source/logs/outputs are preserved in attempt history, and replacements require their own job identity, revision and source-hash audits. Passing the short startup tests supports the repair but does not prove full-run stability, uniquely establish the failure cause, or establish that coarser results are independent of startup settings.

n032_weno9_none: final-hour $h = 195.6$ m; $u_{star} = 0.281$ m/s; surface heat flux = -15.52 W/m²; peak mean $w2 = 0.0927$ m²/s². Between 7–8 and 8–9 h, h changed by -3.7 m and the $w2$ profile changed by 0.00438 m²/s² RMS. Simulation wall time 529.7 s (40114 iterations), excluding preceding compilation/startup.

n032_weno5_none: final-hour $h = 136.1$ m; $u_{star} = 0.242$ m/s; surface heat flux = -11.79 W/m²; peak mean $w2 = 0.05744$ m²/s². Between 7–8 and 8–9 h, h changed by -2.3 m and the $w2$ profile changed by 0.00379 m²/s² RMS. Simulation wall time 505.5 s (39025 iterations), excluding preceding compilation/startup.

n032_weno9_smagorinsky: final-hour $h = 140.6$ m; $u_{star} = 0.223$ m/s; surface heat flux = -9.17 W/m²; peak mean $w2 = 1.12e-06$ m²/s². Between 7–8 and 8–9 h, h changed by 0.1 m and the $w2$ profile changed by $4.74e-08$ m²/s² RMS. Simulation wall time 473.4 s (35312 iterations), excluding preceding compilation/startup.

n064_weno9_none: final-hour $h = 187.8$ m; $u_{star} = 0.279$ m/s; surface heat flux = -15.24 W/m²; peak mean $w2 = 0.09765$ m²/s². Between 7–8 and 8–9 h, h changed by -3.3 m and the $w2$ profile changed by 0.00154 m²/s² RMS. Simulation wall time 1050.0 s (80959 iterations), excluding preceding compilation/startup.

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n064_weno5_none: final-hour $h = 150.9$ m; $u_{star} = 0.248$ m/s; surface heat flux = -12.01 W/m²; peak mean $w2 = 0.06558$ m²/s². Between 7–8 and 8–9 h, h changed by -10.9 m and the $w2$ profile changed by 0.00248 m²/s² RMS. Simulation wall time 1029.6 s (79613 iterations), excluding preceding compilation/startup.

n064_weno9_smagorinsky: final-hour $h = 149.7$ m; $u_{star} = 0.234$ m/s; surface heat flux = -10.16 W/m²; peak mean $w2 = 0.0315$ m²/s². Between 7–8 and 8–9 h, h changed by -21.0 m and the $w2$ profile changed by 0.00986 m²/s² RMS. Simulation wall time 1020.2 s (79350 iterations), excluding preceding compilation/startup.

n128_weno9_none: final-hour $h = 184.4$ m; $u_{star} = 0.270$ m/s; surface heat flux = -13.56 W/m²; peak mean $w2 = 0.1008$ m²/s². Between 7–8 and 8–9 h, h changed by -1.3 m and the $w2$ profile changed by 0.000826 m²/s² RMS. Simulation wall time 4156.8 s (162036 iterations), excluding preceding compilation/startup.

n128_weno5_none: final-hour $h = 162.9$ m; $u_{star} = 0.245$ m/s; surface heat flux = -11.12 W/m²; peak mean $w2 = 0.0696$ m²/s². Between 7–8 and 8–9 h, h changed by 7.9 m and the $w2$ profile changed by 0.00189 m²/s² RMS. Simulation wall time 4058.5 s (160877 iterations), excluding preceding compilation/startup.

n128_weno9_smagorinsky: final-hour $h = 188.1$ m; $u_{star} = 0.277$ m/s; surface heat flux = -13.37 W/m²; peak mean $w2 = 0.08562$ m²/s². Between 7–8 and 8–9 h, h changed by 3.9 m and the $w2$ profile changed by 0.00112 m²/s² RMS. Simulation wall time 4172.6 s (161379 iterations), excluding preceding compilation/startup.

n200_weno9_none: final-hour $h = 178.5$ m; $u_{star} = 0.261$ m/s; surface heat flux = -11.87 W/m²; peak mean $w2 = 0.0993$ m²/s². Between 7–8 and 8–9 h, h changed by 3.3 m and the $w2$ profile changed by 0.000919 m²/s² RMS. Simulation wall time 9396.4 s (253504 iterations), excluding preceding compilation/startup.

n200_weno5_none: final-hour $h = 159.2$ m; $u_{star} = 0.240$ m/s; surface heat flux = -10.04 W/m²; peak mean $w2 = 0.06902$ m²/s². Between 7–8 and 8–9 h, h changed by 1.8 m and the $w2$ profile changed by 0.000677 m²/s² RMS. Simulation wall time 8387.3 s (252093 iterations), excluding preceding compilation/startup.

n200_weno9_smagorinsky: final-hour $h = 188.9$ m; $u_{star} = 0.274$ m/s; surface heat flux = -12.22 W/m²; peak mean $w2 = 0.09298$ m²/s². Between 7–8 and 8–9 h, h changed by 3.3 m and the $w2$ profile changed by 0.00138 m²/s² RMS. Simulation wall time 10660.3 s (252948 iterations), excluding preceding compilation/startup.

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n400_weno9_none: final-hour $h = 165.3$ m; $u_{star} = 0.241$ m/s; surface heat flux = -8.05 W/m²; peak mean $w2 = 0.08926$ m²/s². Between 7–8 and 8–9 h, h changed by 2.4 m and the $w2$ profile changed by 0.000512 m²/s² RMS. Simulation wall time 95437.1 s (510626 iterations), excluding preceding compilation/startup.

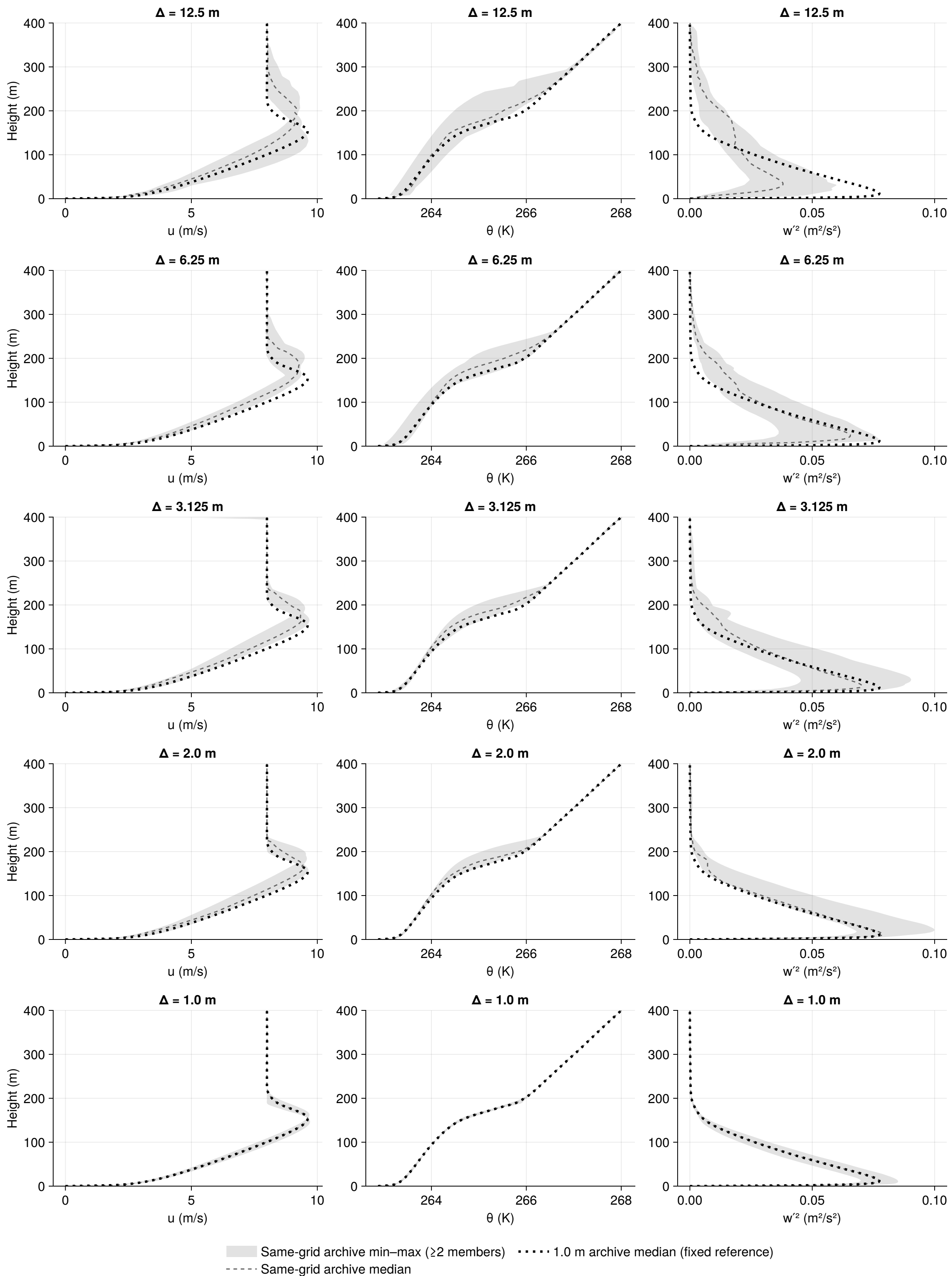
n400_weno5_none: final-hour $h = 159.4$ m; $u_{star} = 0.229$ m/s; surface heat flux = -7.47 W/m²; peak mean $w2 = 0.06807$ m²/s². Between 7–8 and 8–9 h, h changed by 4.7 m and the $w2$ profile changed by 0.000682 m²/s² RMS. Simulation wall time 71341.1 s (506162 iterations), excluding preceding compilation/startup.

Current queue snapshot (job|state|elapsed|start|node or pending reason):

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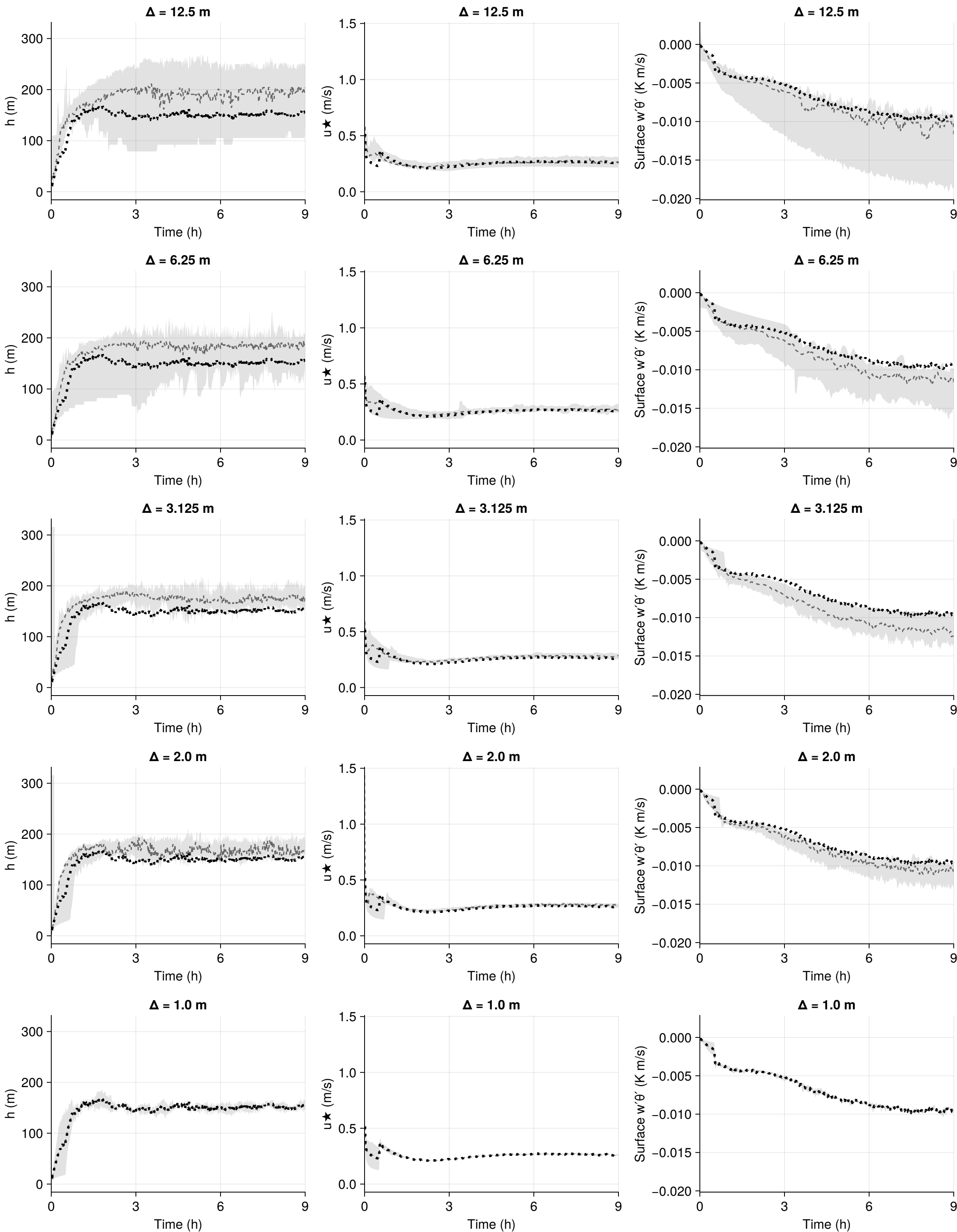
7120_2|RUNNING|8:47:56|2026-09-20T19:37:43|gpu-prod-st-gpu-prod-2

GABLS1 reference archive: final-hour model spread



Archive: nominal final-hour profiles, including some NERSC windows of 474–534 min. Breeze curves, when present: 8–9 h means.
 Native heights interpolated without extrapolation; model membership varies with height and grid. The spread is not observational uncertainty.
 Black dots: fixed 1.0 m ensemble median; interpolation precedes the median. Unavailable references are labeled.

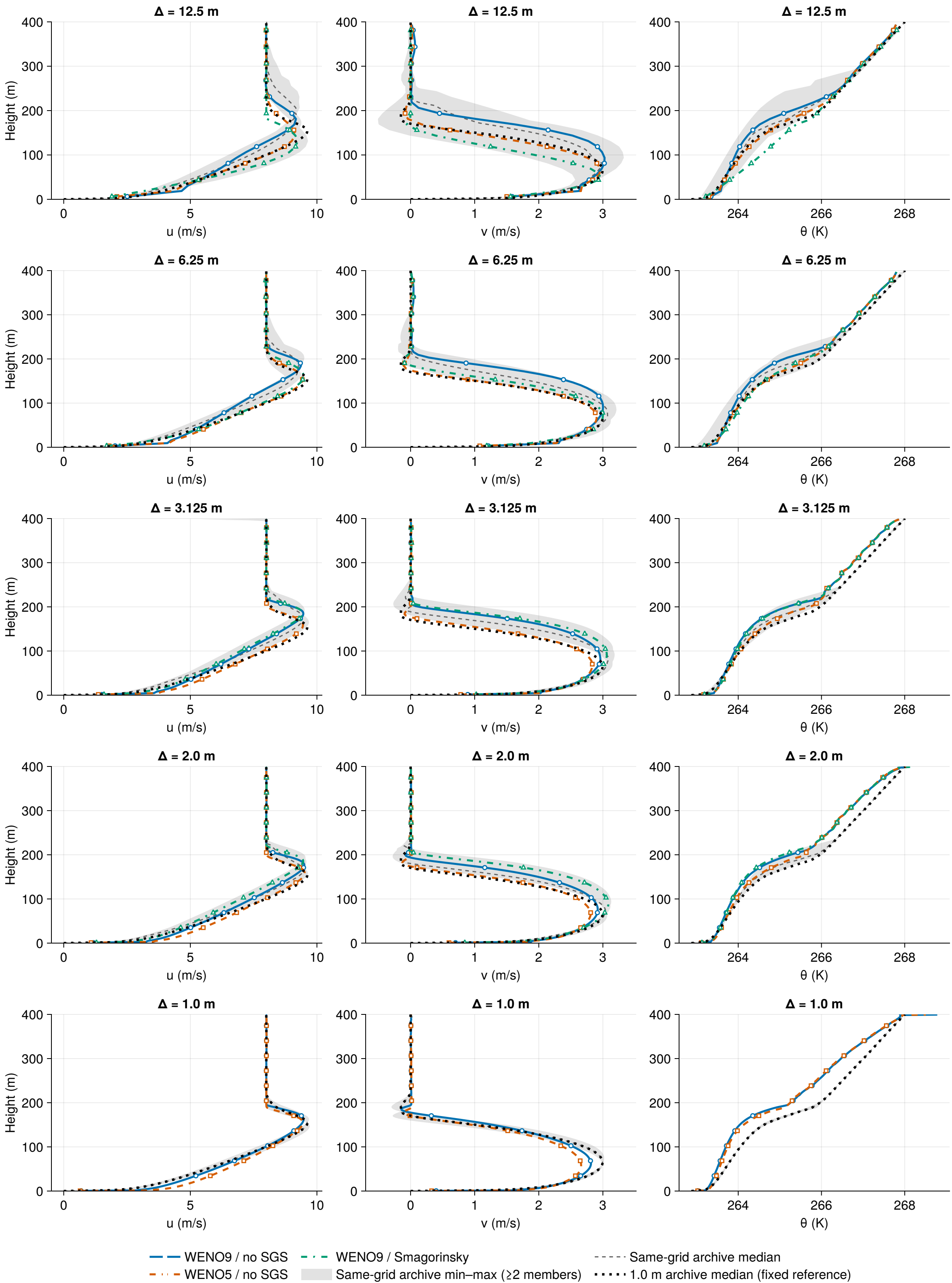
GABLS1 reference archive: evolution



Same-grid archive min-max (≥ 2 members)
 Same-grid archive median
 1.0 m archive median (fixed reference)

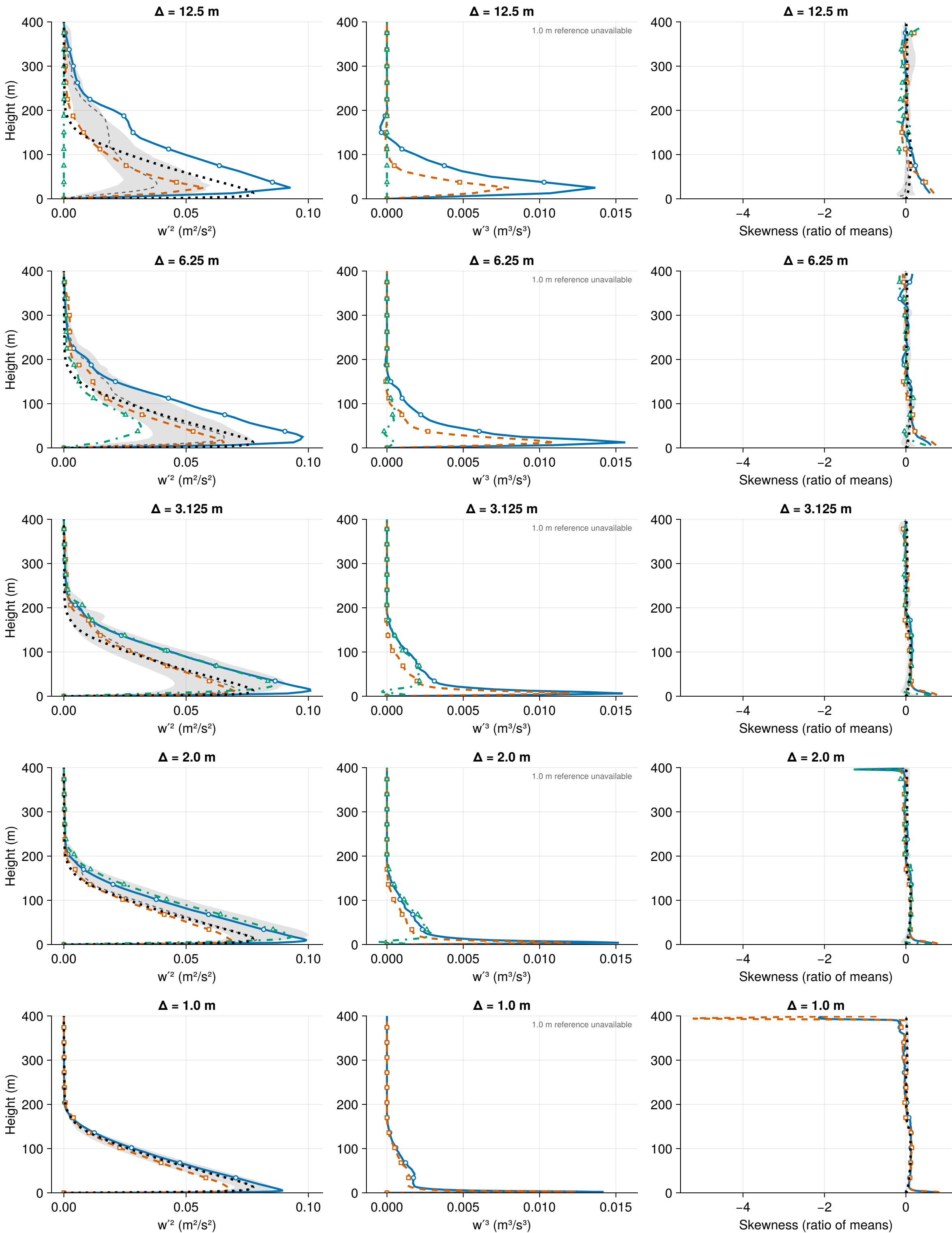
Archive series interpolated to common times without extrapolation; missing values omitted.
 Membership varies with grid and time. Black dots: the same 1.0 m ensemble median in every row. Heat flux is upward positive.

GABLS1: mean wind and temperature



Archive: nominal final-hour profiles, including some NERSC windows of 474–534 min. Breeze curves, when present: 8–9 h means.
Native heights interpolated without extrapolation; model membership varies with height and grid. The spread is not observational uncertainty.
Black dots: fixed 1.0 m ensemble median; interpolation precedes the median. Unavailable references are labeled.

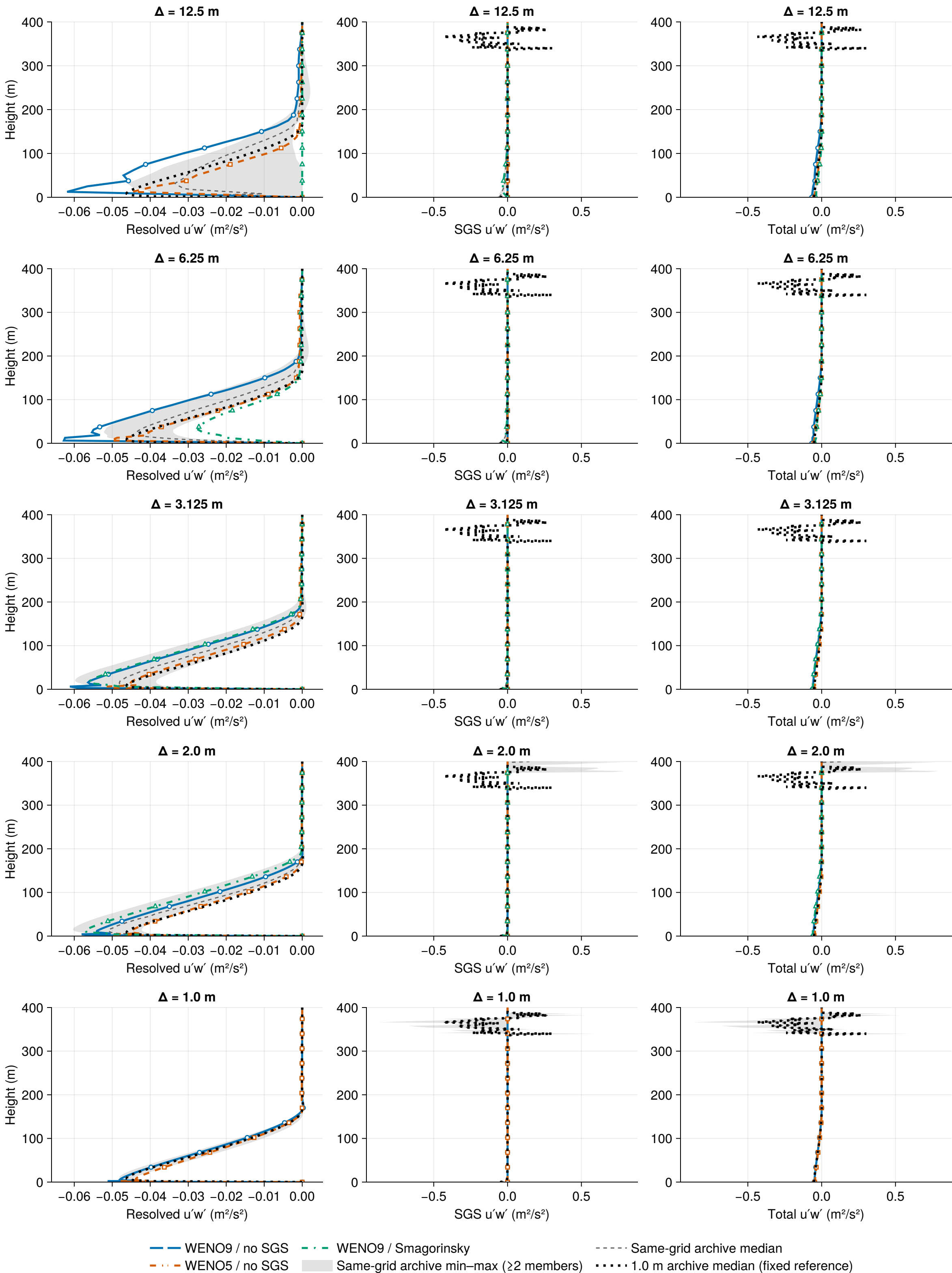
GABLS1: vertical velocity moments



— WENO9 / no SGS - - - WENO9 / Smagorinsky - - - - Same-grid archive median
 - - - - WENO5 / no SGS Same-grid archive min-max (≥ 2 members) ····· 1.0 m archive median (fixed reference)

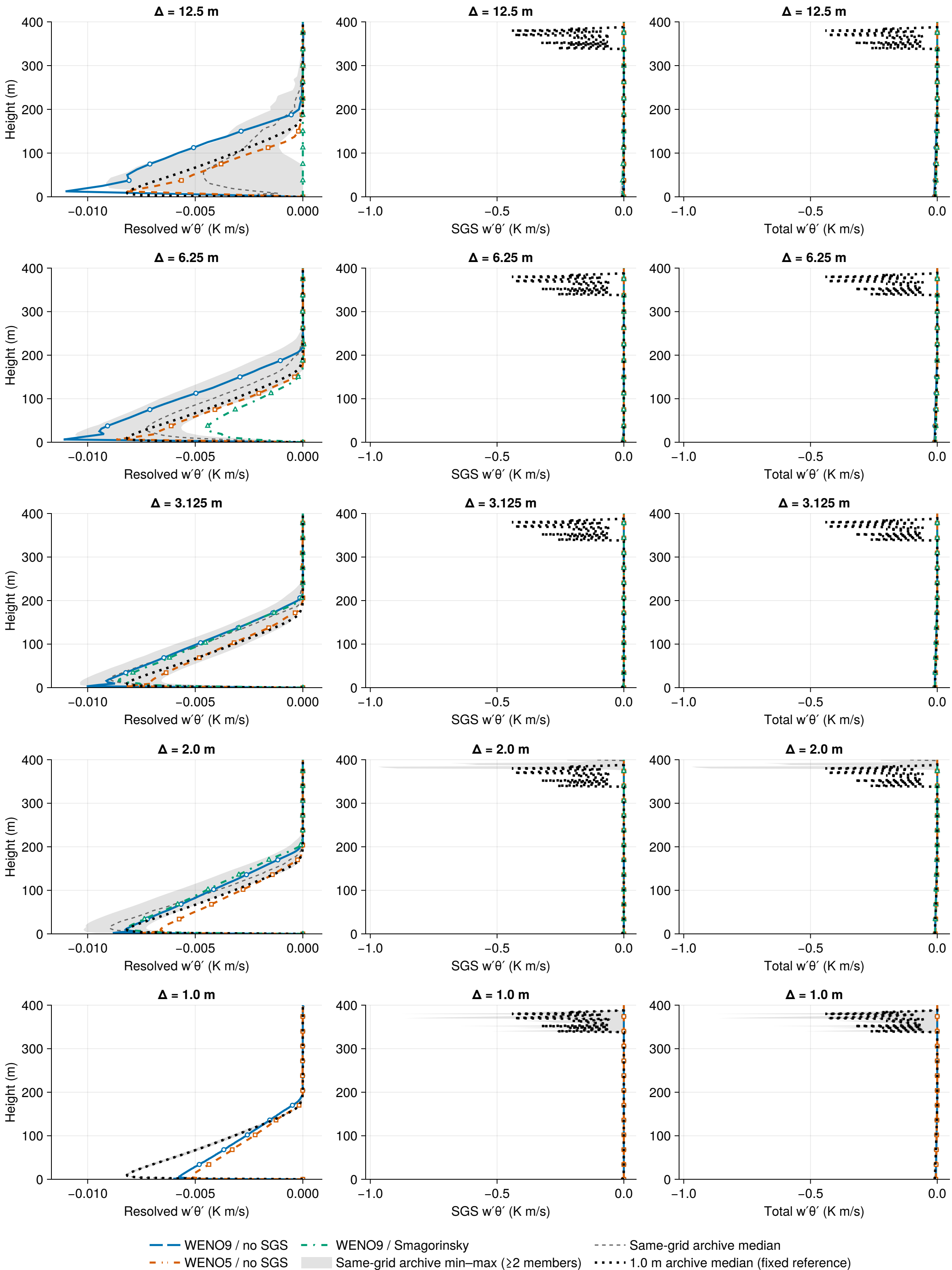
Archive: nominal final-hour profiles, including some NERC windows of 474–534 min. Breeze curves, when present: 8–9 h means. Native heights interpolated without extrapolation; model membership varies with height and grid. The spread is not observational uncertainty. Black dots: fixed 1.0 m ensemble median; interpolation precedes the median. Unavailable references are labeled.

GABLS1: streamwise momentum transport



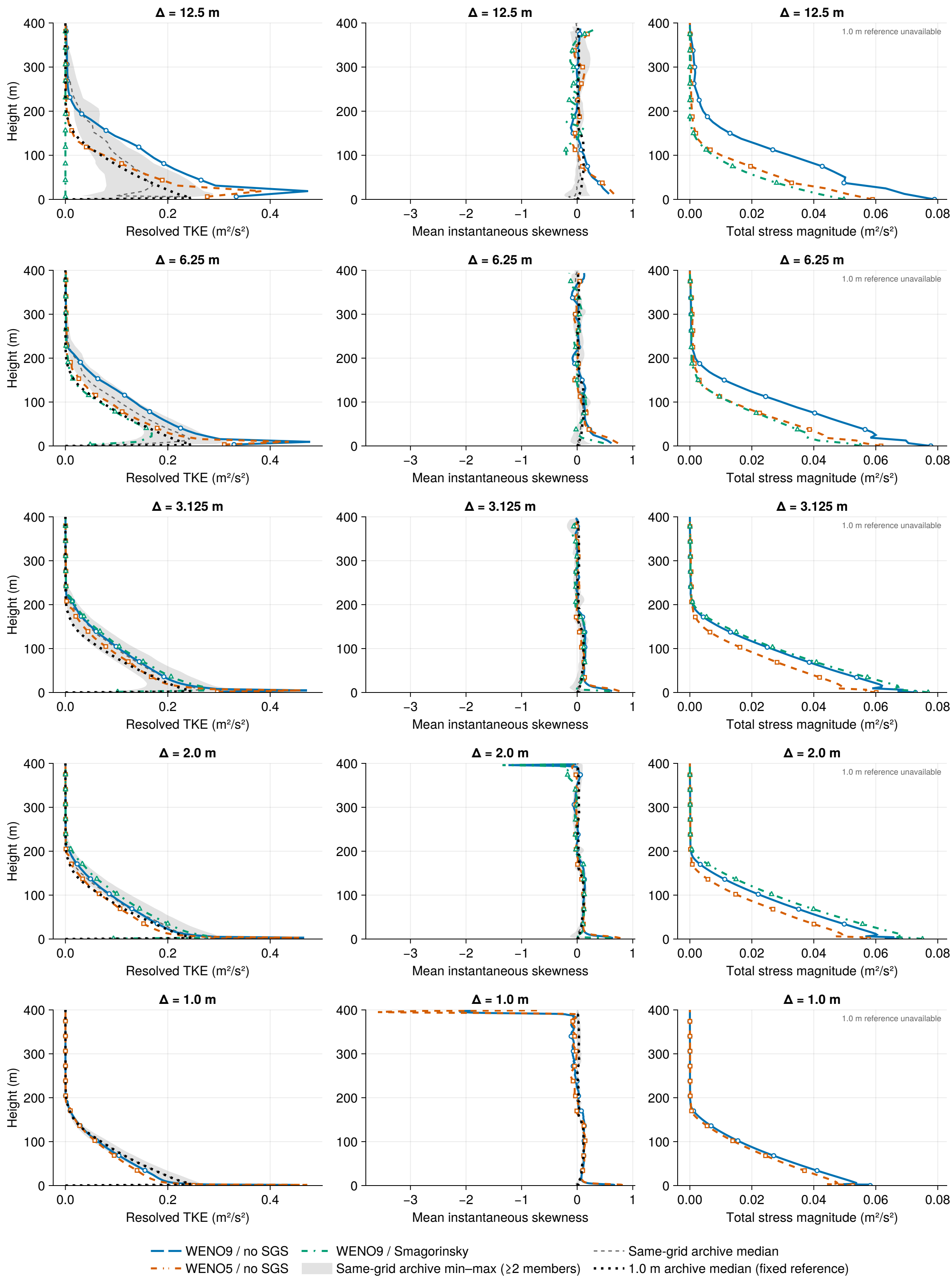
Archive: nominal final-hour profiles, including some NERSC windows of 474–534 min. Breeze curves, when present: 8–9 h means.
 Native heights interpolated without extrapolation; model membership varies with height and grid. The spread is not observational uncertainty.
 Black dots: fixed 1.0 m ensemble median; interpolation precedes the median. Unavailable references are labeled.
 Archive caution: 1.0 m IMUK SGS data contain upper-level anomalies; retained as supplied and flagged in the report.

GABLS1: vertical heat transport



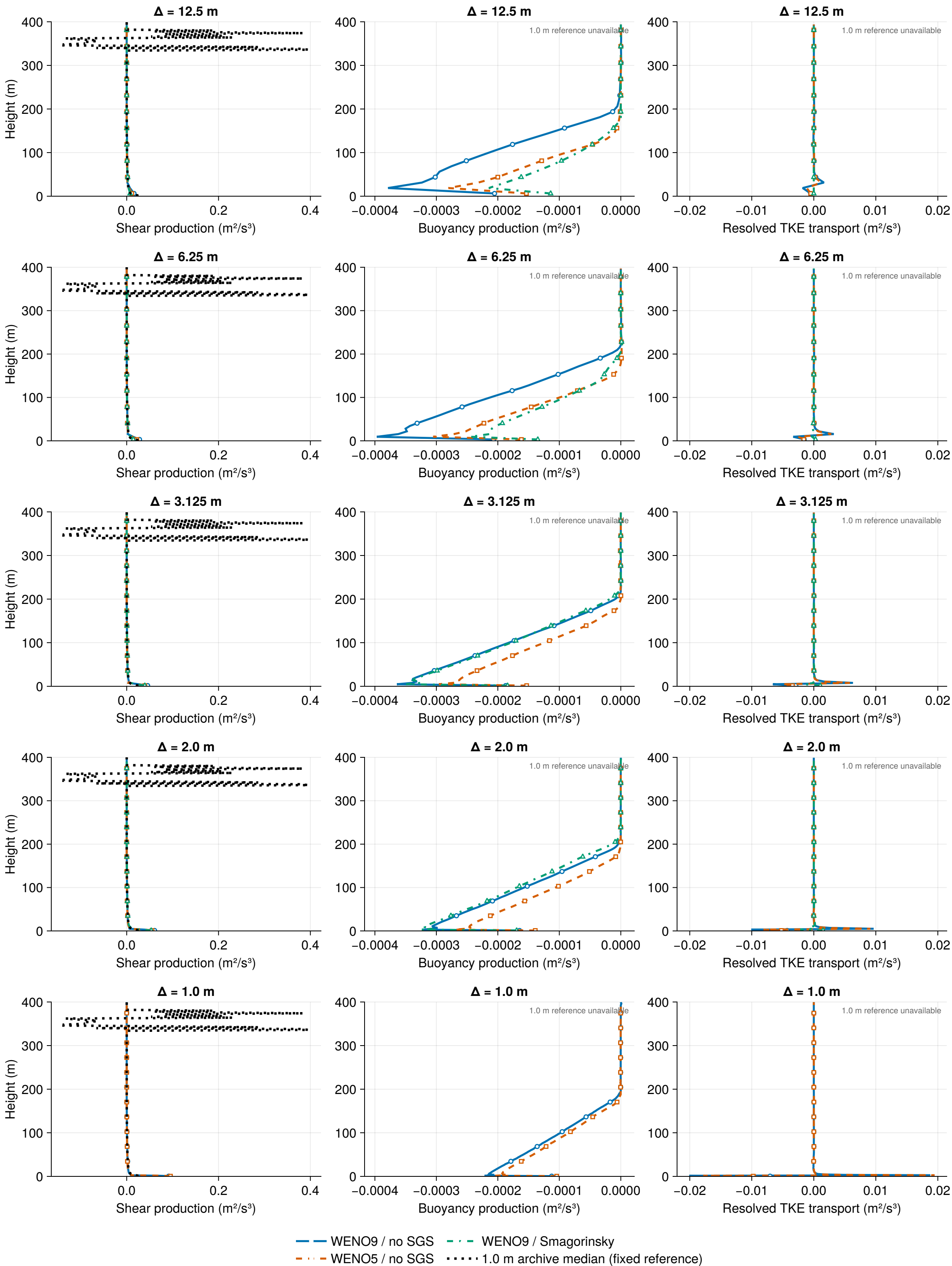
Archive: nominal final-hour profiles, including some NERSC windows of 474–534 min. Breeze curves, when present: 8–9 h means.
 Native heights interpolated without extrapolation; model membership varies with height and grid. The spread is not observational uncertainty.
 Black dots: fixed 1.0 m ensemble median; interpolation precedes the median. Unavailable references are labeled.
 Archive caution: 1.0 m IMUK SGS data contain upper-level anomalies; retained as supplied and flagged in the report.

GABLS1: turbulence intensity, asymmetry and stress



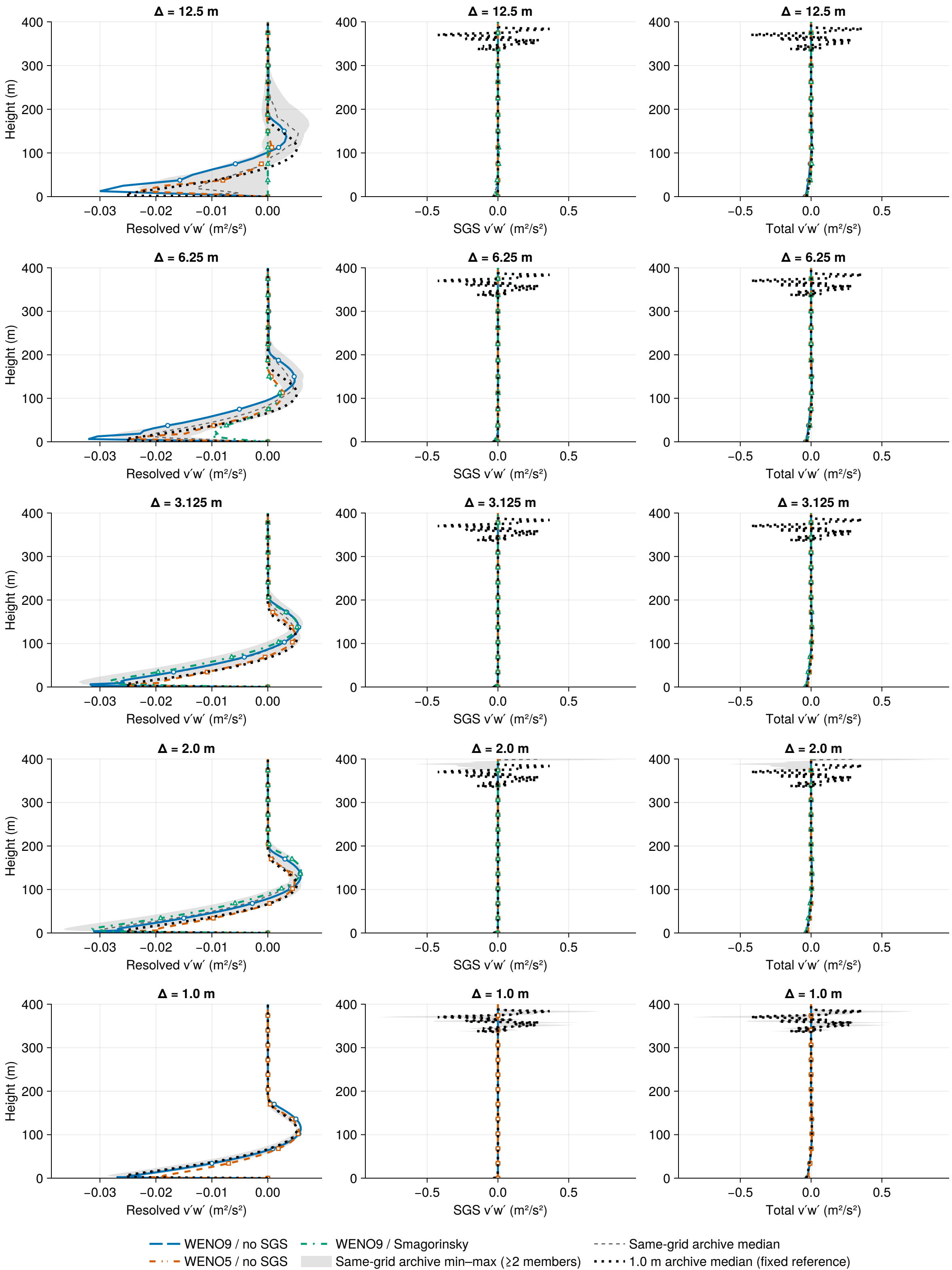
Archive: nominal final-hour profiles, including some NERSC windows of 474–534 min. Breeze curves, when present: 8–9 h means.
 Native heights interpolated without extrapolation; model membership varies with height and grid. The spread is not observational uncertainty.
 Black dots: fixed 1.0 m ensemble median; interpolation precedes the median. Unavailable references are labeled.

GABLS1: selected TKE budget terms



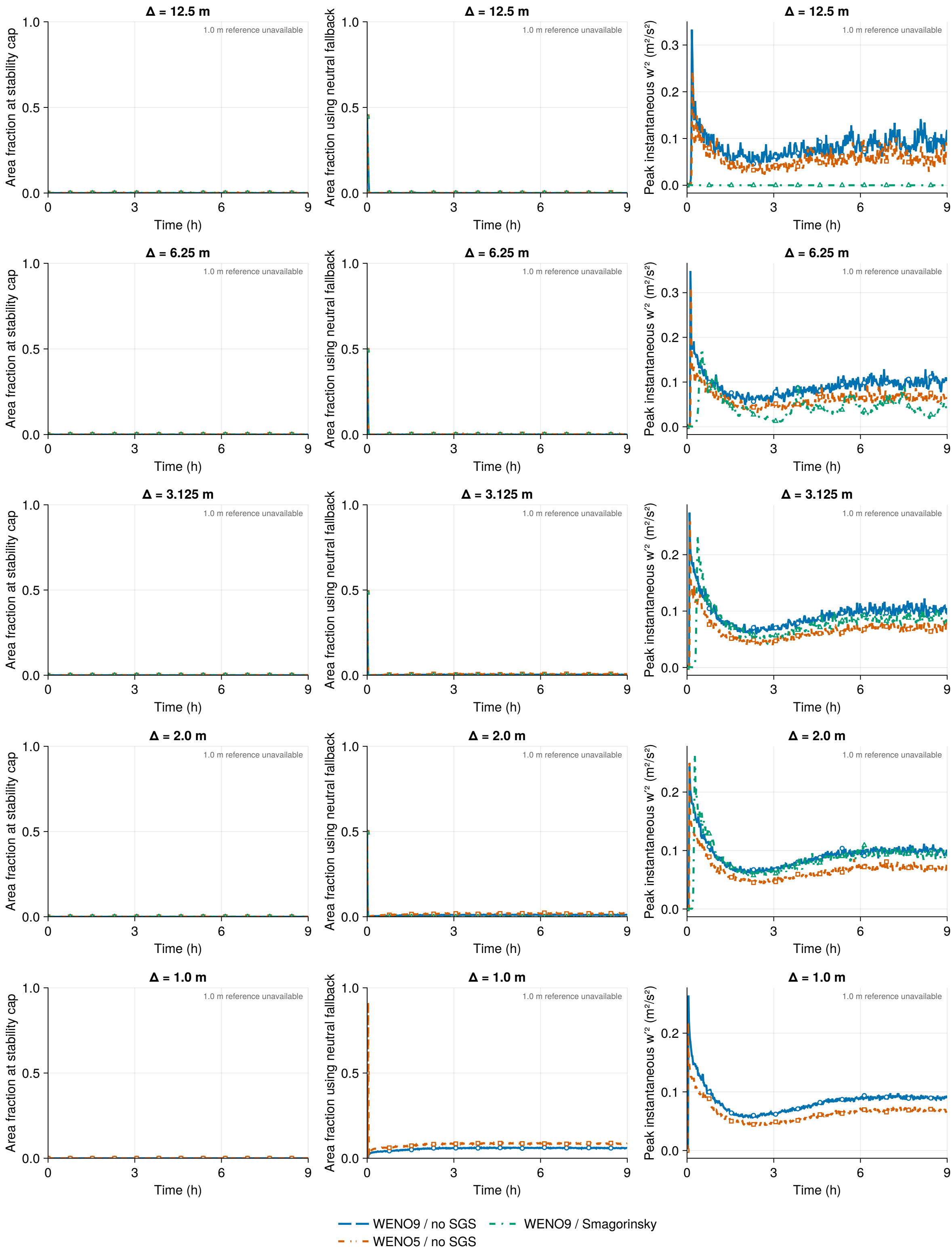
Breeze: 8–9 h means; only fully completed and audited exports. Shown budget terms do not form a closed budget.
A budget residual must not be interpreted as numerical dissipation; unavailable terms are not replaced by zero.
Black dots: fixed 1.0 m ensemble median; interpolation precedes the median. Unavailable references are labeled.
Archive caution: 1.0 m IMUK SGS data contain upper-level anomalies; retained as supplied and flagged in the report.

GABLS1: crosswind momentum transport



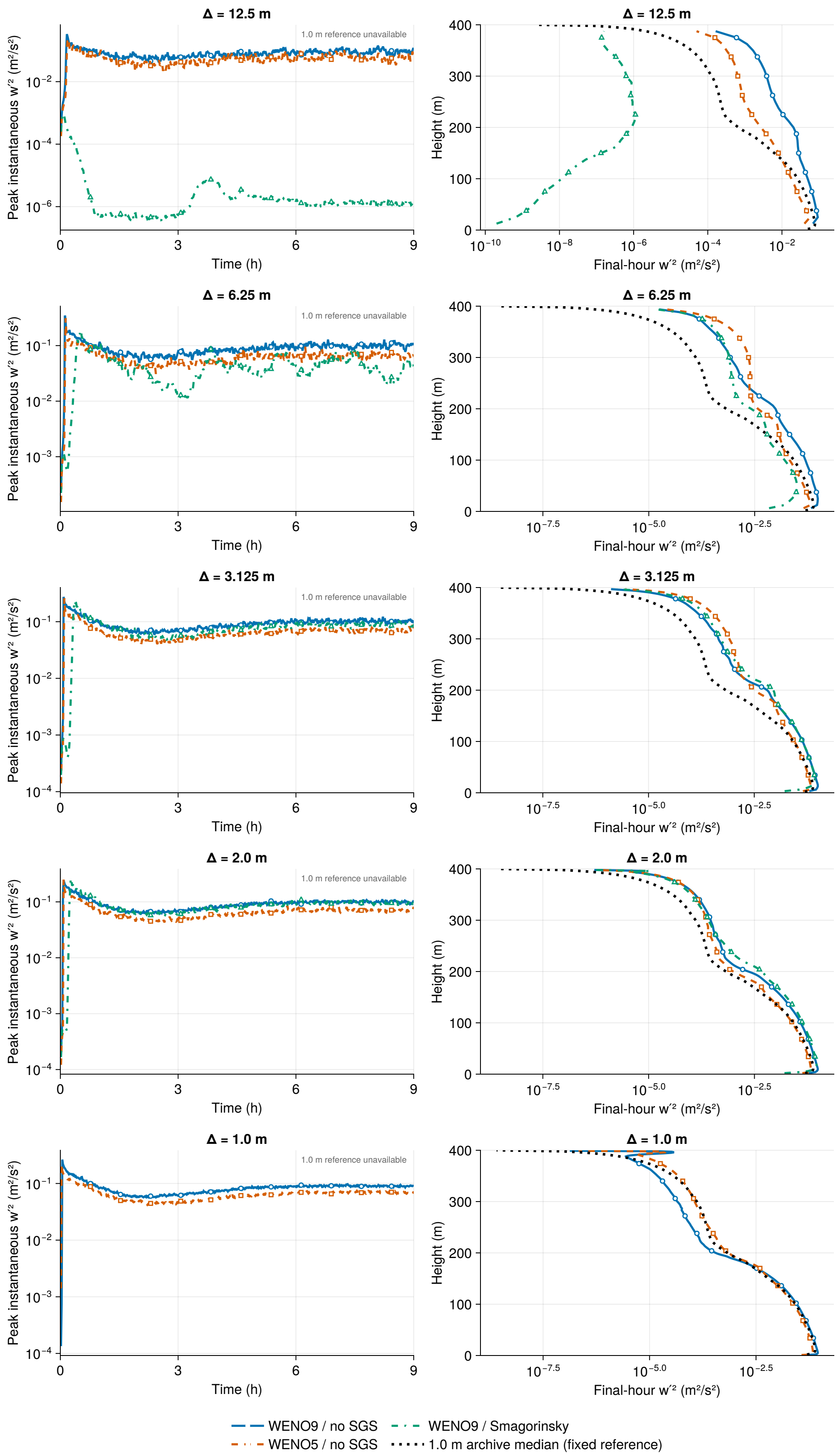
Archive: nominal final-hour profiles, including some NERSC windows of 474–534 min. Breeze curves, when present: 8–9 h means.
Native heights interpolated without extrapolation; model membership varies with height and grid. The spread is not observational uncertainty.
Black dots: fixed 1.0 m ensemble median; interpolation precedes the median. Unavailable references are labeled.
Archive caution: 1.0 m IMUK SGS data contain upper-level anomalies; retained as supplied and flagged in the report.

GABLS1: wall-model regimes and resolved turbulence



Instantaneous 60 s samples. The wall law caps ζ at 10 and uses neutral exchange for nonpositive bulk Richardson number. These fractions identify when wall-model limits participate in the result. No matching archive regime or peak-variance series is available.

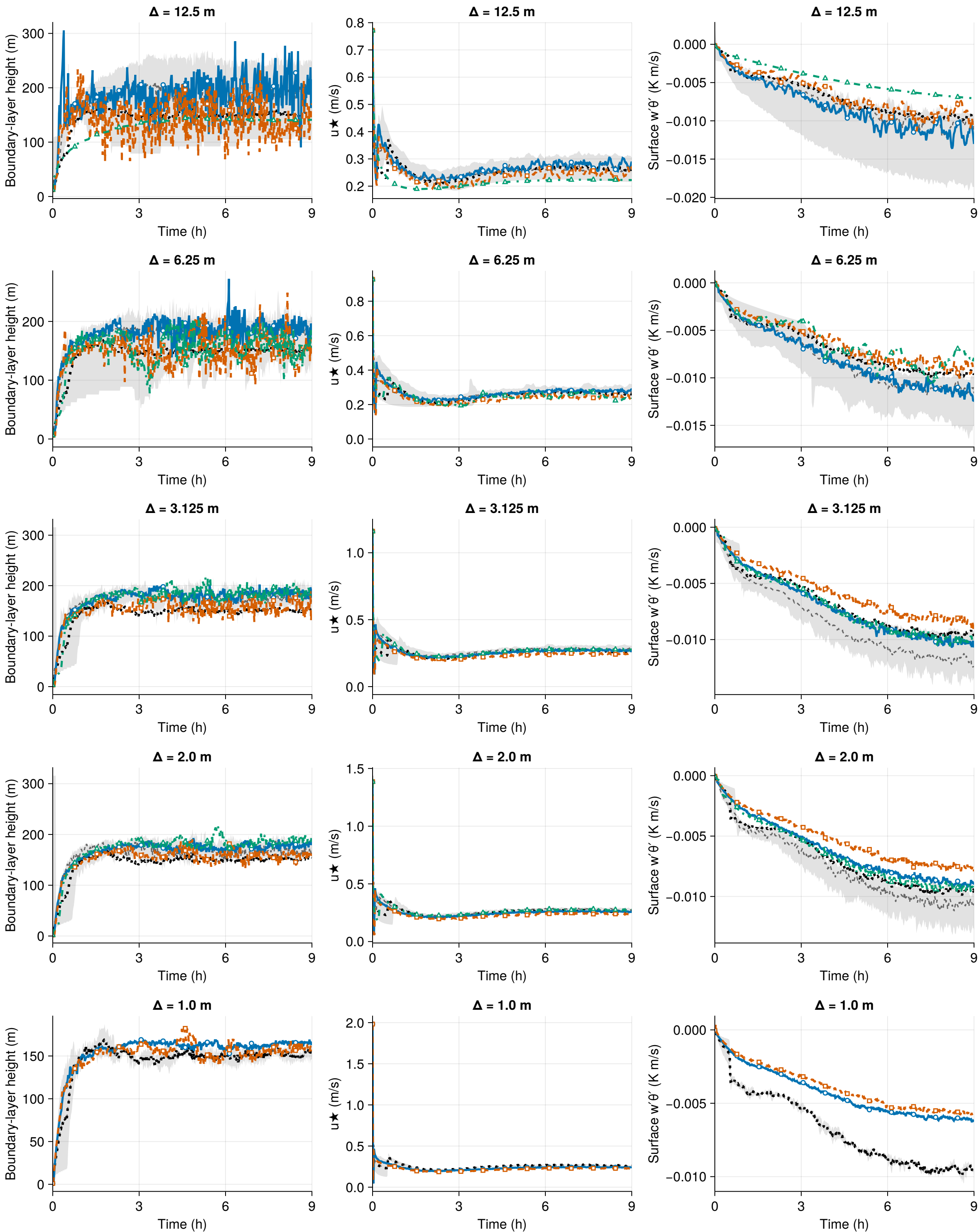
GABLS1: resolved vertical-velocity variance



— WENO9 / no SGS - - WENO9 / Smagorinsky
- . - WENO5 / no SGS . . . 1.0 m archive median (fixed reference)

Logarithmic variance axes distinguish weak resolved motions from sustained turbulence. Left: instantaneous profile maxima; right: 8–9 h means. Black dots: fixed 1.0 m median (right only). No archived peak-variance time series. Nonpositive values omitted without a substituted floor.

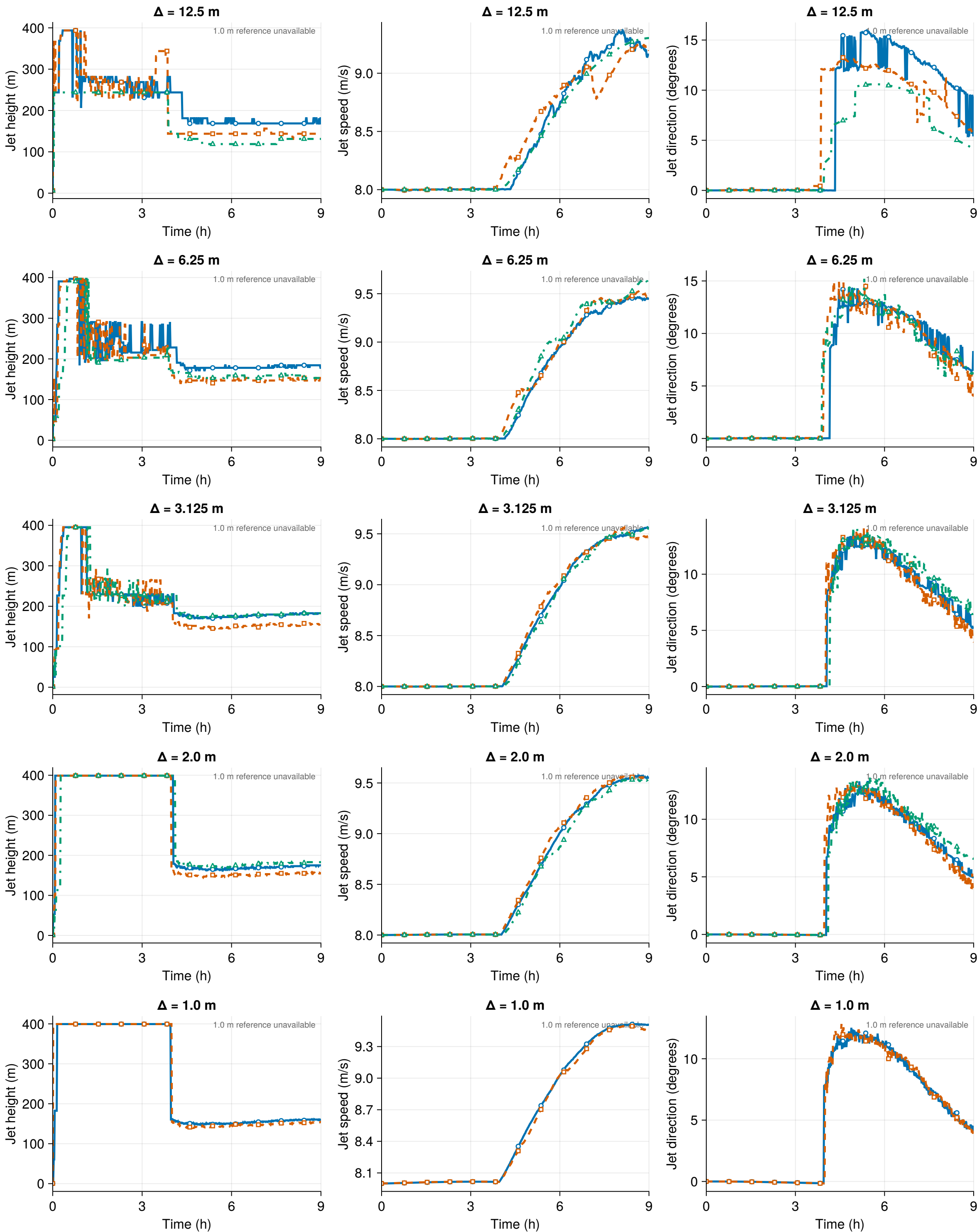
GABLS1: evolution



— WENO9 / no SGS - - - WENO9 / Smagorinsky . . . Same-grid archive median
 - - - WENO5 / no SGS Same-grid archive min-max (≥ 2 members) . . . 1.0 m archive median (fixed reference)

Breeze: instantaneous 60 s samples. Archive series interpolated to common times without extrapolation; model membership varies.
 h = height of 5% total surface stress / 0.95; invalid crossings omitted. Black dots: fixed 1.0 m median. Heat flux is upward positive.

GABLS1: low-level jet evolution



Instantaneous 60 s samples of the maximum horizontal-mean wind speed. Direction is $\text{atan}(v, u)$ relative to eastward geostrophic wind. Invalid jet diagnostics are omitted; a changing peak height can jump between adjacent grid levels. No archived jet time series is available.